

Jan Kwong

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EDUCATION

University of California, San Diego

M.S. in Computer Science (Anticipated). GPA: 3.7.

La Jolla, CA

Dec 2026

University of California, San Diego

B.S. in Computer Science. GPA: 3.8.

La Jolla, CA

Jun 2025

PROJECTS

picoLM: Compact Language Model

Apr 2026 – Jun 2026

- Pretrained a 98M-parameter decoder-only transformer (GQA, SwiGLU, RoPE) in PyTorch on a single shared, interruptible GPU, achieving ~41 perplexity on a 5M-token web-text validation set
- Built a streaming data pipeline over FineWeb, SlimPajama, DCLM, and RedPajama-V2 with on-the-fly hash deduplication and token packing, never materializing the corpus on disk
- Applied knowledge distillation from GPT-Neo-2.7B, reducing perplexity from 70 to 42 by combining teacher-imitation and ground-truth losses

RAG & Context Engineering Pipeline

Apr 2026 – May 2026

- Built a single-turn Q&A Retrieval-Augmented Generation system over a technical OSS documentation corpus, enforcing a strict 2,000-token per-query context budget
- Conducted a systematic sweep of 8+ configurations (varying chunk size, embedding model, retrieval strategy, and reranker settings) to optimize retrieval and generation quality
- Achieved 0.93 Correctness and 1.00 Faithfulness by implementing parent-document retrieval with BM25/vector hybrid search, a BGE cross-encoder reranker, and a dual-citation few-shot prompting strategy

Security Analysis of Apple Intelligence

Oct 2025 – Dec 2025

- Reverse-engineered macOS binaries using Ghidra to map information flow between user input, generative AI models, and local storage
- Identified a privacy vulnerability where the system persisted unencrypted sensitive personal information to local directories, bypassing the ephemeral-only privacy policy
- Developed a Python forensic script to simulate TCC (Transparency, Consent, and Control) inheritance attacks, successfully exfiltrating sensitive user data from sandboxed environments

EXPERIENCE

AI/LLM Engineer

Apr 2026 – Present

QualiFly

Remote

- Built text-type-specific prompts and a shared guardrail for QWriter's LLM writing tutor, routing 6 text types and 4 languages through a title-based type detector so feedback and scoring match each text type
- Built a generate → self-check → repair loop for quiz generation, using structured JSON output, schema validation and prompt versioning to clear stale cached quizzes
- Prototyped a low-temperature LLM grader for Cambridge young-learner writing exams that checks answers against the accepted answers and the official wordlist, choosing it over an embedding-similarity approach

Graduate Teaching Assistant

Apr 2026 – Present

UC San Diego

La Jolla, CA

- Leads weekly discussion sections for Intro to Research and Neurobiology of Cognition mentoring sections of ~80 students through empirical study design, statistical validation, and core neuroscience concepts
- Designs weekly review slides and multiple-choice practice questions aligned with lecture content, holds weekly office hours, and gives students concrete feedback on assessments to help them grasp difficult ideas and improve their work

Undergraduate Research Assistant

Jan 2024 – Dec 2024

UC San Diego LASR Lab

La Jolla, CA

- Co-authored a paper (accepted for publication on JASA) benchmarking OpenAI Whisper against 75 human transcribers on 300 English sentences from 20 diverse accented speakers
- Automated WER computation with Python scripts, discovering that humans significantly outperform models on isolated words, suggesting potential limitations in models' training data or acoustic context requirements

SKILLS

ML & AI: PyTorch, Hugging Face, Scikit-learn, AI Engineering, Machine Learning, NLP, Computer Vision, Data Engineering, Data Analytics, LLM Pretraining, Knowledge Distillation, RAG

Programming Languages: Python, SQL, Java, C, C++, JavaScript, HTML, CSS, ARM, Clojure, Rocq

Tools & Infrastructure: Pandas, NumPy, Git, Claude Cowork, Docker, Kubernetes, Ghidra, CI/CD

CS Foundations: Data Structures, Algorithms, Database Systems, Operating Systems, Computer Architecture, Computer Security, Agile

Languages: Cantonese (Native), English (Native), Mandarin (Native), Japanese (Professional)